Neel Jagdishchandra Prajapati

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.55
Shree Saraswati Vidhyalaya	Class XII (Gujarat Secondary & Higher Secondary Education Board, Gandhinagar)	2023	73.85%
DRGD Sarvajanik Vidhyalaya	Class X (Gujarat Secondary & Higher Secondary Education Board, Gandhinagar)	2021	96.00%

SKILLS

Technical Skils

Python, C++, SQL, JavaScript

Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, DBMS, Computer Networks

PostgreSQL, Data Pipelines, Streaming Systems, Vector Databases

AI/ML

PyTorch, TensorFlow, Scikit-learn, NLP, OpenCV, Machine Learning, RAG, LLMs, LangChain, LangGraph

AWS, Docker, Kubernetes, Terraform, Git, CI/CD, Linux

Jira, Streamlit, Google Colab

PROJECTS

Agentic RAG System | Python, Streamlit, ChromaDB, Gemini API | GitHub

- Designed and built a modular Retrieval-Augmented Generation pipeline with independently testable stages (ingestion, chunking, embedding, retrieval, and generation), using ChromaDB for persistent vector storage with a JSON-based fallback for restricted environments.
- Implemented hallucination-control mechanisms -- similarity-threshold filtering, context-only prompting, zero-temperature generation, and a fixed "unknown" response for unanswerable queries -- to ensure grounded, source-cited answers.
- Built an interactive Streamlit interface for document upload, querying, and answer inspection, with an extractive fallback answerer for use without an GEMINI API key.

Telco Customer Churn Prediction & Deployment | Python, XGBoost, FastAPI, Docker, AWS ECS, MLflow, GitHub Actions | GitHub

- Built an end-to-end churn prediction pipeline -- feature engineering, XGBoost model training, and threshold tuning -- achieving 0.84 ROC-AUC and 0.82 recall on held-out test data, with experiments tracked via MLflow.
- Deployed the model as a REST API (FastAPI) and interactive Gradio UI, containerized with Docker and hosted on AWS ECS Fargate behind an Application Load Balancer.
- Set up a CI/CD pipeline with GitHub Actions to automatically build, push, and redeploy the containerized service on every code change, resolving networking and health-check issues along the way (security groups, ALB target routing).

Financial Anomaly Detection Pipeline | Python, MLflow, FastAPI, Docker, Streamlit | GitHub

- Built an end-to-end MLOps pipeline to detect fraudulent credit-card transactions, with data validation, preprocessing, and class-weighted baseline models tracked via MLflow.
- Deployed a FastAPI inference service in Docker, validated through a CI pipeline of linting, tests, and build checks.
- Added PSI-based drift monitoring with an automated retraining gate and a Streamlit dashboard for live predictions; achieved 0.90 precision and 0.80 PR-AUC.

SCHOLASTIC ACHIEVEMENTS

- Cleared **NTSE Stage 1** (state) & **Stage 2** (National) in 2021 ;

Selected as one of **2000** *Students Nationwide by NCERT*

POSITIONS OF RESPONSIBILITY

- Students Representative *(BUDGET HEAD)* | **NSS IIT HYD** - Leading social initiatives for 3+ years, impacting 1000+ students. Organized large-scale events like Clean India Drives, orphanage & School visits and Awareness sessions .

- Inbound Coordinator | **International Relations Cell** - Hosted IIT Hyderabad globally; facilitated collaborations with Japanese firms, professors, and companies to connect with the student community

EXTRACURRICULAR

- Organized **Department Day**, coordinating events for professors and students across UG, PG, and PhD levels.

- **UG Researcher**, Propulsion Systems - Working on design and manufacturing aspects of a **Rotating Detonation Engine** to enhance propulsion efficiency.